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Working Memory Capacity and Individual Differences in the Making of Reinstatement and Elaborative Inferences

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This study investigated the role of working memory capacity on the making of reinstatement and causal elaborative inferences during the reading of natural texts. In order to determine participants' working memory capacity, they were asked to take the reading span task before they took part in the study. Those participants that were identified as high or low working memory capacity readers were asked to perform a lexical decision task in two conditions: pre-inference and inference. In the pre-inference condition, target words representing reinstatement or causal elaborative inferences were presented immediately before the sentences that were predicted to prompt them. In the inference condition, the target words were presented immediately after the sentences that were predicted to prompt the inferences. Results indicated that, for the high working memory capacity readers, lexical decision times were faster at the inference compared to the pre-inference locations for both types of inferences. In the case of low working capacity readers, lexical decision times were faster at the inference compared to the pre-inference locations only for reinstatement inferences. These findings suggest that working memory capacity plays a role in the making of causal inferences during the comprehension of natural texts.
Keywords: working memory capacity, causal elaborative inferences, reinstatement inferences.

Este estudio investigó el rol de la capacidad de memoria de trabajo en la realización de inferencias de restablecimiento y elaborativas durante la lectura de textos naturales. Los participantes realizaron la tarea de amplitud lectora, y aquellos que fueron identificados como lectores de alta o baja capacidad de memoria de trabajo realizaron una tarea de decisión léxica en dos condiciones. En la condición de pre-inferencia, las palabras de prueba se presentaron inmediatamente antes de la oración que promovía la realización de la inferencia. En la condición de inferencia, las palabras de prueba se presentaron inmediatamente después de la oración que promovía la inferencia. Los resultados indicaron que los lectores de alta capacidad de memoria de trabajo tenían menores tiempos de reacción en la condición de inferencia en comparación con la condición de pre-inferencia para ambos tipos de inferencias. Los lectores de baja capacidad de memoria de trabajo tenían tiempos de reacción menores en la condición de inferencia en comparación con la condición de pre-inferencia, solo en el caso de las inferencias de restablecimiento. Estos hallazgos sugieren que la capacidad de memoria de trabajo tiene un rol en la realización de inferencias causales durante la comprensión de textos naturales.
Palabras clave: capacidad de memoria de trabajo, inferencias causales elaborativas, inferencias repositivas.

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Text comprehension involves the construction of a coherent mental representation, which requires for the reader to build bridges between the new information and his or her background knowledge through the making of inferences (van den Broek, 1990; 1994). Several factors influence whether inferences are made or not: contextual support, distance between the sentences that need to be linked, conceptual overlap between antecedent and target sentence, individual differences (Cook, Gueraud, Was, & O'Brien, 2007; Gerrig & O'Brien, 2005; Guéraud, Harmon, & Peracchi, 2005; Guéraud, Tapiero, & O'Brien, 2008; Lassonde & O'Brien, 2009; Magliano & Millis, 2003; Mc Namara & Magliano, 2009; McNamara & McDaniel, 2004; McNamara & O'Reilly, 2009; McNamara & Scott, 2001; Millis, Magliano, & Todaro, 2006; O'Brien, Cook, & Gueraud, 2010; O'Reilly & McNamara, 2007; Ozuru, Dempsey, & McNamara, 2009; Peracchi & O'Brien, 2004; Harmon-Vukic, Guéraud, Lassonde, & O'Brien, 2009) Since the 90's, there has been considerable evidence that working memory is one of the cognitive factors that account for individual differences in the making of such inferences (Carpenter, Just, & Shell, 1990; Estévez & Calvo 2000; Friedman & Miyake, 2004; Calvo 2001, 2004; Linderholm et al., 2000; Linderholm & van den Broek, 2002; McNamara, de Vega, & O'Reilly, 2007; Virtue, van den Broek, & Linderholm, 2006).

van den Broek (1994) identifies different types of causal inferences. Connective inferences are made when the reader identifies a causal relation between the statement that he or she is reading, and information that remained activate in working memory after processing the immediately previous sentence. Reinstatement inferences are made when the reader reactivates information presented previously in the text, in order to maintain sufficient causal justification for the statement that he or she is reading. Causal elaborative inferences draw on the readers´ background knowledge to identify a causal antecedent that is not explicitly mentioned in the text. Predictive inferences draw on the readers' background knowledge to anticipate upcoming text events.

Working memory refers to the mechanisms and processes involved in the control, regulation and active execution of complex cognitive activities, and the active maintenance of relevant information (Miyake & Shah, 1999). According to Just and Carpenter (1992) and Just, Carpenter, & Keller (1996), the capacity of this mechanism reflects the maximum amount of activation available to perform cognitive functions. This amount varies among individuals, and is typically assessed by complex working memory span tasks, which require for the participant to read or listen to a number of sentences (sometimes also to perform a grammatical or semantic verification task), in order to remember the last word of each sentence (Daneman & Carpenter, 1980). This measure has been linked to general language comprehension (Just & Carpenter, 1992).

Studies on the role of working memory capacity in the making of causal inferences have focused on connective inferences (Singer, Andrusiak, Reisdorf, & Black, 1992; Singer & Ritchot, 1996) and predictive inferences (Calvo, 2001; 2004; Estévez & Calvo 2000, Linderholm, 2002).

Singer et al. (1992) and Singer and Ritchot (1996) examined the making of connective inferences. Singer et al. (1992) asked participants to read pairs of causally connected sentences (for example: *'Julia took an aspirin. Her headache went away'*), and varied the number of sentences that intervened between them. In order to determine the occurrence of the inference, participants had to answer an inference question after reading the last sentence (e.g., *did the aspirin take Julia's headache away?*). Results indicated that, when the number of sentences between the causally connected pair was high, subjects with high working memory capacity were more accurate than those with low working memory capacity. There was no difference between the high and low working memory capacity participants when there were no intervening sentences between the pair. Consistent with these results, Singer and Richot (1996) found that participants with high working memory capacity had faster answer times to questions about the world knowledge that needed to be activated to make the connective inference (for example, in the pair '*Julia took an aspirin. Her headache went away',* readers need to activate the knowledge that '*Aspirin relieves headaches'*) than low working memory capacity readers. In other words, it seems that high working memory capacity readers are more capable of making connective inferences than low working memory capacity participants when the causally connected sentences are distant in the text.

Estévez and Calvo (2000), Calvo (2001; 2004) and Linderholm (2002) examined the role of working memory capacity in the making of predictive inferences. Estévez and Calvo (2000) asked participants to read sentences that were predictive or non-predictive of an event, followed by words that either represented the predicted event (confirming words) or an inconsistent event (disconfirming words). Participants' task was to name these words. Results indicated that, 1000 msec after reading the last word of the context sentences, high working memory capacity participants had shorter naming latencies for the confirming words that followed the predicting sentences than the control sentences, and longer naming latencies for the disconfirming target words that followed the predicting relative to the control sentences. There were no naming latencies differences for the low working memory capacity participants. After 1500 msec, there were faster naming responses following the predicting relative to the control context in both working memory capacity groups, and no difference for the disconfirming words. These results suggest that high working memory capacity readers make predictive inferences earlier than low working memory capacity readers. In another study, Calvo (2001) monitored eye fixations as participants read context sentences that predicted likely events (e.g., *'Three days before the examination the pupil went to the library, looked for a separate table and opened his notebook'*) or non-predicting control sentences, followed by continuation sentences in which a target word represented either the event to be inferred ('*The pupil studied for an hour approximately'*) or an unlikely event ('*The pupil slept for an hour'*). Results indicated that high working memory capacity readers spent less time reading the last region of the continuation sentences (e.g., '*approximately'*), and looking back from it in the predicting relative to the control condition. Calvo (2004) also monitored eye fixations as participants' read context sentences ('*When the party was over, there were bags and papers all over the floor'*) followed by continuation sentences ('*Susan swept the floor thoroughly*') that confirmed the inferences suggested. Once more, high working memory capacity readers spent less time reading the final word of the continuation sentences ('*thoroughly'*), and looking back from them. In other words, predicting contexts seem to facilitate the making of predictive inferences for high working memory capacity readers.

Linderholm (2002) also examined the making of predictive inferences, but focusing on the causal structure of the text. She asked participants to read two-sentence texts containing low, moderate and high causal sufficiency (that is, events that suggested a low, moderate and high likelihood that a particular consequence would occur), and to name target words representing the predictive inferences presented 250 or 500 msec after the texts. Results indicated faster naming times for the high working memory capacity participants at both time delays in the high causal sufficiency condition, and no differences indicating that low working memory capacity participants had made the predictive inference in any of the conditions. In other words, high working memory capacity readers, but not low working memory capacity readers, make predictive inferences when texts contain high causal sufficiency.

To sum up, there have been several studies suggesting a role for working memory capacity in the making of connective and predictive elaborative inferences during the reading of experimental texts (Calvo, 2001, 2005; Estévez & Calvo 2000; Linderholm, 2002; Singer et al., 1992; Singer & Ritchot, 1996). Yet, it would be interesting to run more studies to investigate the role of this capacity in the making of other types of inferences, such as reinstatements and backward elaborative inferences, and in the processing of natual texts. That is, studies on connective inferences have found a role for working memory capacity, but mainly when text demands increase (Calvo, 2001). For example, Singer et al. (1992) found that this capacity facilitated the making of inferences when the causal antecedent and the target sentence were distant, but not when there were no intervening sentences between them. It would then be interesting to examine the role of working memory capacity when readers are required to make reinstatement inferences with a small number of intervening sentences between antecedent and target sentence. Studies on causal elaborative inferences have tended to focus on predictive inferences. That is, inferences about "*what will happen next*" in the text. It is then not as clear what the role of working memory capacity is on the making of backward elaborative inferences, or inferences that involve the activation of background knowledge to identify the cause for the event that is being described in the text (van den Broek, 1990; 1994). It has also been suggested that it would be important to study comprehension by using long and natural discourse (Graesser, Magliano, & Haberlandt, 1994). That is, the short and artificial texts that have tended to be used in comprehension studies might be less interesting and coherent than natural texts, and not reflect the processes that take place when readers face them. The purpose of this study was to address these issues, by focusing on the role of working memory capacity in the making of reinstatement and causal backward elaborative inferences during the reading of natural texts (fairy tales). In order to examine this, we asked high and low working memory capacity participants (as assessed by the *reading span task*) to read three popular fables, and to perform a lexical decision task in two conditions: pre-inference and inference. In the pre-inference condition, participants were presented with target words representing the reinstatement or causal elaborative inferences immediately before the sentences that were predicted to prompt them. In the inference condition, the target words were presented immediately after the sentences that were predicted to prompt the inferences. We expected high working memory capacity readers to make both types of inferences. That is, given that they have more cognitive resources available for storage and processing than low working memory capacity readers, they should be able to reactivate previously read sentences and world knowledge that is necessary to obtain causal justification for the sentence that they are reading. And, given that they have fewer cognitive resources available to make inferences, low working memory capacity readers should make reinstatement but not causal elaborative inferences. That is, they should be able to reinstate causally connected information from the prior text, given that this necessary for making the text coherent and for comprehension (Calvo, 2001; van den Broek, 1990; 1994). And, they should not be able to activate background knowledge that is not mentioned in the text, given that it is not necessary for comprehension, but rather to refine the explicit information in the message (Calvo, 2001).

Method

Participants

Sixty-four undergraduate students volunteered to participate in the study for course credit. Their mean age was 21 (*SD* = 2.5). The sample was drawn from introductory psychology classes.

Materials

Materials consisted of three fairy tales: '*Jorinde and Joringel',* '*Rapunzel'* and '*Rumpelstiltskin',* compiled by Grimm and Grimm (1812/2000). Their extension was approximately 40 sentences, and they were presented in Spanish. Each tale was analyzed independently by two judges following the causal network model procedures proposed by Trabasso and Sperry (1985), in order to identify those sentences that required the realization of reinstatement and causal elaborative inferences. Differences between the raters were resolved through discussion. On the basis of these judgments, two sentences that required the making of reinstatement and two sentences that required the making of elaborative inferences were identified for each tale (see Appendix). A target word describing the inference was identified by the two judges. Following van den Broek, Rohleder, and Narvaez (1996), in the case of the reinstatement inferences the target words were selected from the causal antecedent sentences. In the case of the elaborative inferences, the words were agreed upon the two judges and compared to the judgment of a third rater. The mean distance between the causal antecedent sentences and the target statements was 1.67 (SD = .52) Three extra locations were selected for the presentation of pseudowords. These locations were selected at random, as long as they were at least two sentences away from the sentences selected for the presentation of target words. Thus, in total each text contained two words that constituted reinstatements, two words that constituted elaborative inferences, and three letter strings that were non-words.

Instruments

In order to determine individual differences in working memory capacity, participants took the reading span task (Daneman & Carpenter, 1980), which has been validated and adapted to Argentine Spanish in a study by Barreyro, Burin, and Duarte (2009). This task requires for participants to read out aloud a set of unrelated sentences in the computer screen (one sentence at a time), and to recall the final word of each sentence when the set is over. The sentence sets start at two sentences, and increase up to five sentences per set. When the participant successfully recalls at least two out of the three trials that comprise each sentence set level, he or she moves to the next level (which involves one more sentence, and one more word to recall). The study by Barreyro et al. showed that those participants that obtain a Span of 2 or recall 6 words fall in the 25th percentile, while those that obtain scores above a Span of 3.5 and recall more than 24 words fall above the 75th percentile. This task was administered to 126 participants. Sixty-four of them were selected: 32 were identified as low working memory capacity readers (their score was 2 or less), and 32 were identified as high working memory capacity readers (their score was 3.5 or more).

Design and Procedure

Participants were tested in one individual session. They were first administered the reading span task. Those that were identified as high or low working memory capacity readers (lower or upper quartile), proceeded to the reading task. This task took approximately 30 minutes. Participants read each sentence at their own pace on the computer screen, and pressed the '+' key when they were ready to move to the next sentence. At the points selected for testing, they saw a row of asterisks instead of the following sentence, and had to perform the lexical decision task. That is, a strings of letters appeared on the screen, and they had to respond yes or no to whether it constituted a word, by pressing the YES or NO keys in a button box. The experiment was run using MEL 2.0 experimental software (Schneider, 1988).

Reaction times were measured in two conditions: pre-inference and inference, for the low and high working memory capacity readers. In the pre-inference condition, participants were presented with the target word representing the reinstatement or causal elaborative inference immediately before the sentence that was predicted to prompt it. In the inference condition, the target words were presented immediately after the sentences that were predicted to prompt the inferences. Each subject received half the target words in the pre-inference condition, and half in the inference condition. In each of the working memory capacity groups (high or low), each participant was randomly assigned to receive the first target word in the inference or pre-inference condition, the second in the pre-inference or inference condition, and so on. In total, each subject was presented with 12 target words: 6 representing reinstatements and 6 representing elaborative inferences; and with 9 pseudowords.

Results

Lexical decision times were submitted to a 2 x 2 x 2 mixed factors analysis of variance treating subjects (F_1, t_1) and items (F_2, t_2) as the random variable. Working memory capacity (low or high), probe location (pre-inference or inference) and type of inference (reinstatement or elaborative) were the independent variables. Working memory capacity was a between-subjects variable, and probe location and type of inference were within-subjects variables. An alpha level of .05 was used to determine significance for all analyses. Responses that were more than 2 standard deviations from the mean reaction time for a particular word were excluded from the analysis, and replaced by the word's mean reaction time for that condition. Table 1 shows mean response times for the low and high working memory capacity readers in each probe location condition (pre-inference and inference), for both

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Table 1
Mean reaction times as a function of target location (pre-inference or inference), inference type (reinstatement or elaborative) and working memory capacity (high or low)

		Working Memory Capacity			
		High		Low	
		M	*SD*	*M*	*SD*
Reinstatement	Pre-inference	770	249	1016	337
	Inference	698	211	913	294
Causal elaborative	Pre-inference	759	269	867	246
	Inference	677	204	880	281

types of inferences (reinstatements and elaborations). Only correct responses were included in the analyses. The percentage of lexical decision errors was 2.34 %.

There was a three-way interaction among the three variables $F_1(1, 62) = 7.26$, $MSE = 8512.79$, $p < .01$, $\eta^2 = .11$, $F_2(1, 10) = 6.15$, $MSE = 6319.86$, $p < .05$, $\eta^2 = .38$. Follow-up analyses indicated that, for the high working memory capacity readers, there was no interaction between probe location and type of inference $F_1(1, 31) = .141$, $MSE = 4872.57$, $p =.71$, $\eta^2 = .00$, F_2 $(1, 10) = .10$, $MSE = 738.38$, $p < .76$, $\eta^2 = .01$.; and there was no difference in reaction times according to type of inference, F_1 $(1, 31) = .86$, $MSE = 9972.53$, $p = .36$, $\eta^2 = .03$, F_2 $(1, 10) = .04$, $MSE = 10610.42$, $p < .85$, $\eta^2 = .00$. There was a significant difference in reaction times according to probe location, F_1 $(1, 31) = 17.16$, $MSE = 11007.53$, $p <.001$, $\eta^2 = .36$, $F_2(1, 10) = 65.14$ $MSE = 738.38$, $p < .001$, $\eta^2 = .87$ for both types of inferences. That is, subjects responded faster to target words in the inference compared to the pre-inference condition in the case of reinstatement inferences, $t_1(31) = 3.30$, $p < .01$, $t_2(5) = 6.21$; $p < .01$ and causal elaborative inferences, t_1 $(31) = 3.60$, $p < .001$, $t_2(5) = 5.37$; $p <$,01. According to these results, high working memory capacity readers do not differ in their reaction times to reinstatement and elaborative inferences, and they make both types of inferences.

For low working memory capacity readers, there was an interaction between probe location and type of inference $F_1(1, 31) = 8.71$, $MSE = 12513.01$, $p < .01$, $\eta^2 = .22$, $F_2(1, 10) = 6.79$, $MSE = 10759.43$, $p < .05$, $\eta^2 = .40$. Further analyses indicated that, in the case of reinstatement inferences reaction times were significantly faster in the inference compared to the pre-inference condition, $t_1(31) = 2.99$, $p < .01$, $t_2(5) = 2.80$; $p < .05$, suggesting that low working memory capacity readers make this type of inference. In the case of elaborative inferences, there was no significant difference in reaction times between the pre-inference and inference conditions, t_1 $(31) = .47$, $p = .64$, t_2 $(5) = 1.36$; $p < .23$, suggesting that low working memory capacity readers do no make this type of inferences.

Discussion

Since the 90's, there has been converging evidence that individual differences in working memory capacity have a role in the comprehension of texts (Calvo, 2001; 2004; Estévez & Calvo 2000; Just & Carpenter, 1992; Linderholm, 2002; Singer et al., 1992; Singer & Ritchot, 1996) The purpose of this study was to explore such role, by examining the influence of verbal working memory on the making of causal inferences.

According to Just and Carpenter's capacity constrained comprehension theory (1992), reading comprehension is mediated by working memory capacity. That is, by the maximum amount of activation that each individual has available for the processing and storage of information. During the reading of texts, the role of working memory is both to maintain information from previous reading cycles active in order to connect it to the new reading cycles, and to activate the readers' background knowledge to integrate it to the mental representation that is being constructed. Individual differences in working memory capacity are expected to constrain some of these processes, and to result in qualitative and quantitative differences in text comprehension by low and high working memory capacity readers (Daneman & Carpenter, 1980; Friedman & Miyake, 2004).

Prior research has found a role for working memory capacity in the making of bridging and predictive inferences (Calvo, 2001; 2005; Estévez & Calvo 2000, Linderholm, 2002; Singer et al., 1992; Singer & Ritchot, 1996). The purpose of this study was to extend this research, by examining the role of working memory capacity in the making of reinstatement and causal elaborative inferences, and by using natural texts as materials. Reinstatement inferences are made when the reader identifies a causal connection between the sentence that he or she is reading, and information presented in previous sentences. Causal elaborative inferences are made when the reader activates background knowledge in order to obtain sufficient causal justification for the sentence that he or she is reading. Given

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that the antecedent has been presented previously, reinstatement inferences are expected to be made easily. The making of elaborative inferences is expected to be more demanding, given that they require for the reader to activate knowledge that is not mentioned explicitly in the text (Estevez & Calvo, 2000).

In order to explore these issues, we asked participants to perform a lexical decision task in two conditions: pre-inference and inference. In the pre-inference condition, target words were presented immediately before the locations that were predicted to elicit the inferences. In the inference condition, target words were presented at the locations that were predicted to elicit the inferences. We found that high working memory capacity readers had faster response times in the inference compared to the pre-inference condition for both types of inferences. Low working memory capacity readers had faster response times in the inference compared to the pre-inference condition only for reinstatement inferences. In other words, high working memory capacity readers seem to make reinstatement and elaborative inferences as they read natural texts, but low working memory capacity readers seem to only make reinstatement inferences. Considering these findings, we can suggest that high working memory capacity readers have resources available to process the text, reinstate previously presented information, and activate background knowledge necessary to comprehend the sentence that they are reading. It, thus, appears that these readers not only benefit from the information that the text provides, but are also able to connect it to their previous knowledge, allowing them to construct a rich mental representation On the other hand, low working memory capacity readers seem to be able to reinstate previously read information when it is necessary to obtain causal justification for the sentence they are reading, but they are not able to activate the background knowledge necessary to make elaborative inferences. It seems that their cognitive resources are being used to process the text, leaving them with no resources to activate background knowledge. Given this limitation, these readers will not connect the information provided by the text with their previous knowledge, and will not construct a rich text representation.

Considering the results of this study, we can propose that it extends prior research on the role of individual differences in the comprehension of texts. It explores the role of working memory capacity in the making of reinstatement and causal elaborative inferences, and it focuses on the processing of natural texts. The use of natural text materials is important, given that it has been proposed that experimenter created texts might not reflect the processes that take place during the reading of non-artificial texts (Graesser et al., 1994). By approaching the comprehension of such materials, we are able to extend some of the conclusions that have been reached with artificial texts about the importance of working memory capacity to the comprehension of natural texts.

Interesting questions for future research are related to the role of individual differences in working memory capacity in the making of other types of inferences during the processing of other types of texts. It will be interesting to examine whether individual differences will play the same role, or if other variables will play a more important role.

In conclusion, findings from this study extend previous work on individual differences in the comprehension of texts, by approaching the making of causal inferences during the comprehension of natural texts, and by providing evidence that working memory capacity has a role in the establishment of causal connections between the information that the text provides and the readers' background knowledge.

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APPENDIX

Excerpt of the fairy tale ′*Rapunzel*′ and sample target words

1. There were once a man and a woman who had long in vain wished for a child.
2. Finally, the woman got pregnant.
3. These people had a little window at the back of their house from which a garden that belonged to an enchantress could be seen, where no one dared to go into.
4. One day the woman was standing by the window, and looking down into the garden, when she saw a bed which was planted with the most beautiful rampion, and she had a craving for it.
   TARGET WORD: *pregnant* (REINSTATEMENT INFERENCE)
5. She told her husband "if I can't get some of the rampion, I shall die."
6. In the twilight of the evening, he clambered down over the wall, hastily clutched a handful of rampion, and took it to his wife.
   TARGET WORD: *satisfy* (ELABORATIVE INFERENCE).
7. The next day she longed for it three times as much as before.
8. The husband let himself down again; but when he had clambered down the wall he saw the enchantress standing before him, yelling angrily at him.

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